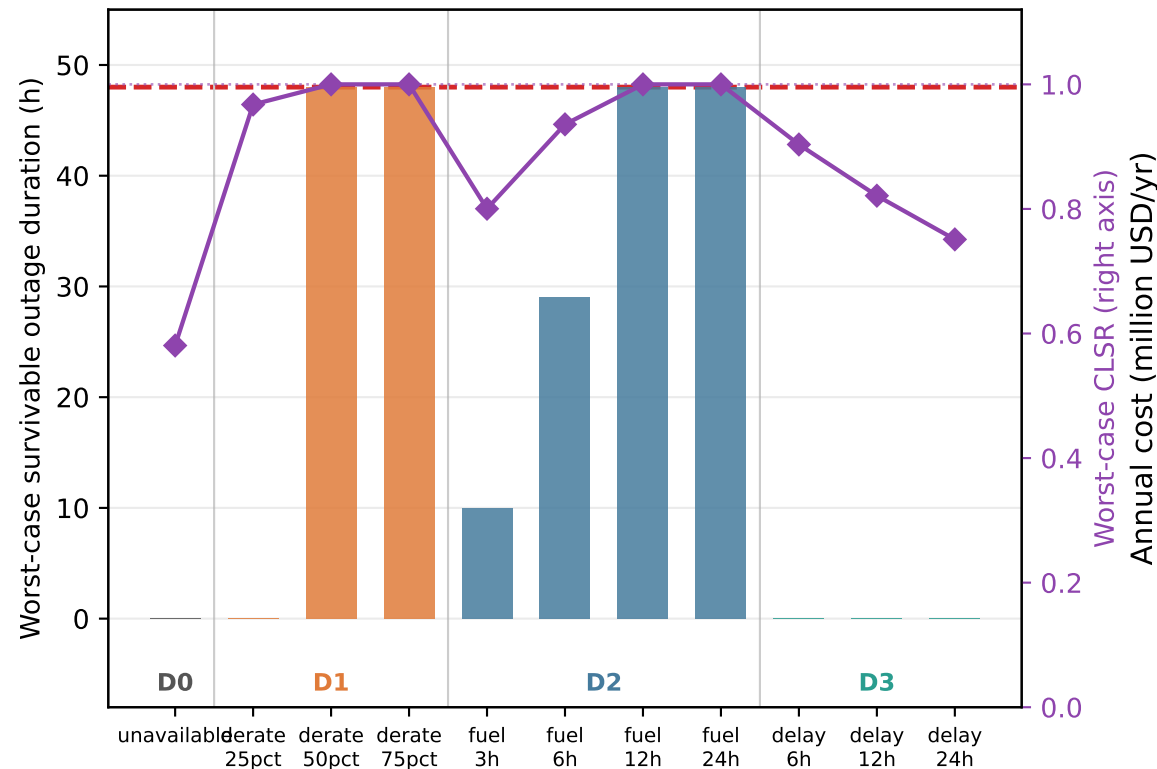
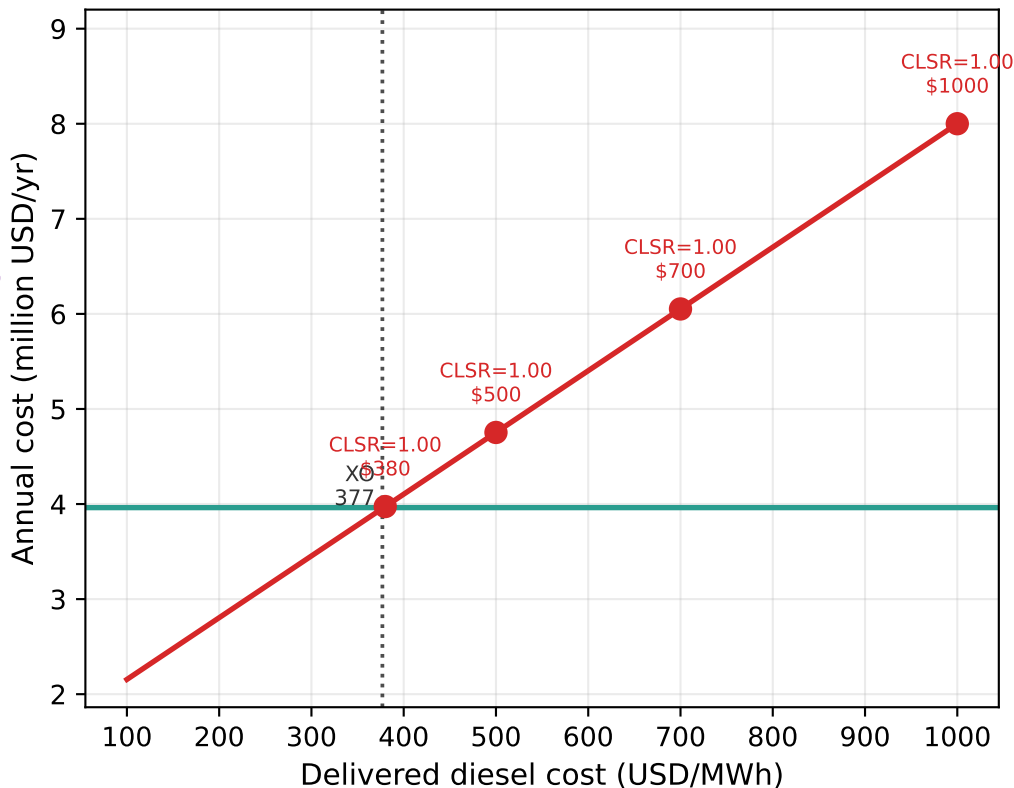


Diesel-Outage Robustness: Graded Support vs H₂ Reference

D0-D3 Diesel Availability — Resilience Impact
(diesel_battery, 10 seeds, 48 h/4 events/yr; H₂ always 48 h)



D4 Diesel Available — Economic Crossover
(carbon=150 USD/tCO₂; CLSR=1.0 for both above crossover)



D0 unavailable
 D2 fuel budget
 H₂ reference 48 h
 D1 derated
 D3 delayed
 DB CLSR_min
 Diesel+battery (D4: diesel avail.)
 H₂+battery (carbon=150)
 Crossover ~377 USD/MWh